

EROL AKARSU, Ph.D.

Lead Machine Learning Architect | Principal Software Engineer | AI Solutions Innovator

eakarsu@gmail.com Richmond, Virginia, USA linkedin.com/in/eakarsu 107+ Projects Portfolio

PROFESSIONAL SUMMARY

Visionary technology leader with **25+ years** of experience architecting enterprise-scale software solutions. Ph.D. in Computer Science from Syracuse University with deep expertise in Machine Learning, Generative AI, and distributed systems. Currently leading ML/AI initiatives at Lowe's Companies, developing transformer-based image curation systems and GenAI assistants serving millions of users.

Built 50+ production AI applications spanning Machine Learning, NLP, Computer Vision, and Generative AI across 15+ industries.

PROFESSIONAL EXPERIENCE

Lead Machine Learning Architect Lowe's Companies, Inc.

Jan 2024 - Present

- Lead development of ML transformer models for optimal image selection from vast product databases
- Architect Generative AI-powered Project Quota Assistance system for real-time Q&A
- Design and implement LLM solutions for home improvement project assistance

Principal Software Engineer Lowe's Companies, Inc.

Oct 2020 - Jan 2024

- Engineered pivotal Lowe's.com features: free delivery, expedited delivery, promotions
- Spearheaded Apache Beam data pipelines and Elastic Search frameworks
- Served as single-threaded leader between senior VPs and engineering teams

Senior Software Architect Wipro Limited (Lowe's Account)

May 2018 - Oct 2020

- Built enterprise search on GCP using Terraform, Packer, Ansible, Jenkins
- Reduced product indexing time by 9x (360 to 40 minutes for 5M products)
- Architected blockchain products with Web3, Solidity, NextJS

Master Software Engineer Capital One

Aug 2016 - Dec 2017

- Led developer.capitalone.com overhaul using Spring, NodeJS, AngularJS
- Automated API testing via MVC, Swagger, FSM, and Graph Computing

Independent Software Consultant US Government & Enterprise Clients

2003 - 2017

- Built Turkish search engine using Nutch/SOLR/Hadoop with 15 ranking criteria
- Achieved 99.5% extraction rate for 700K lines of COBOL code via XQUERY

Software Engineer Cisco Systems

Nov 1999 - Jul 2007

- Developed unified communications software solutions for enterprise clients

TECHNICAL SKILLS

AI & MACHINE LEARNING

LLMs/GPT Transformers GenAI PyTorch
TensorFlow NLP Computer Vision RAG

BIG DATA & SEARCH

Apache Spark Apache Beam Elasticsearch Solr
Kafka

CLOUD & DEVOPS

GCP AWS Kubernetes Docker Terraform

LANGUAGES & FRAMEWORKS

Python Java Scala TypeScript React
Node.js

DATABASES

PostgreSQL MongoDB Oracle Redis

EDUCATION

Ph.D., Computer Science Syracuse University

1994-1999 | GPA: 3.8

M.S., Computer Science Syracuse University

1994-1996 | GPA: 3.7

M.S., Computer Engineering Istanbul Technical University

1991-1993

B.S., Computer Engineering Ege University

1986-1991

CERTIFICATIONS

Deep Learning Specialization - Coursera
Neural Networks & Deep Learning
Advanced ML Algorithms

AI Application Portfolio Overview

Comprehensive collection of production-ready AI applications demonstrating full-stack development expertise

50+

AI APPLICATIONS

107

TOTAL PROJECTS

15+

INDUSTRIES

Full

STACK

2024

LATEST TECH

Business & Finance AI 12 Projects



AI Churn Prediction System
CUSTOMER ANALYTICS

SaaS for predicting customer churn with AI risk scores, segments, and intervention recommendations.

React

Node.js

PostgreSQL

Claude



AI Pricing Optimizer
REVENUE OPTIMIZATION

Dynamic pricing with competitor tracking, demand analysis, confidence scoring.

React

Express

PostgreSQL



AI Financial Reports
FINANCIAL ANALYTICS

Automated reporting with AI insights, trend analysis, KPI dashboards, PDF export.

React

Node.js

PostgreSQL



AI Due Diligence
M&A / INVESTMENT

M&A platform with document analysis, risk assessment, automated reporting.

React

Node.js

OpenRouter



AI Inventory Forecasting
SUPPLY CHAIN

Demand forecasting, supplier scoring, restock recommendations, anomaly detection.

React

Express

Claude AI



AI Tax Preparation
ACCOUNTING

Document parsing, deduction optimization, compliance checking, form generation.

React

Node.js

OpenRouter



AI Competitive Analysis
MARKET INTELLIGENCE

Competitor tracking, SWOT analysis, market trend monitoring, strategic recommendations.

React

Express

Claude AI



AI Regulatory Compliance
COMPLIANCE

Regulation tracking, audit preparation, risk assessment, automated reporting.

React

Node.js

Claude AI



AI Business Automation
PROCESS AUTOMATION

Workflow design, task automation, approval chains, optimization suggestions.

React

Node.js

Workflow



AI Procurement
SUPPLY CHAIN

Vendor management, RFQ automation, contract analysis, supplier recommendations.

React

Node.js

OpenRouter



AI Finance Platform
FINANCIAL SERVICES

Portfolio management, transaction analysis, AI-powered financial planning.

React

Node.js

JWT Auth



AI Data Analyst
BUSINESS INTELLIGENCE

Natural language queries, automated insight generation, visualization creation.

React

Node.js

Recharts



AI Marketing Copy

CONTENT MARKETING

AI copywriting for marketing content, ad copy, emails with brand voice customization.

React

Node.js

Claude AI



AI Content Studio

CONTENT CREATION

Content management with AI writing, calendar, collaboration, multi-channel publishing.

React

Node.js

Rich Text



AI Email Triage

EMAIL AUTOMATION

Intelligent email management with categorization, priority scoring, sentiment analysis.

React

Express

NLP



AI Review Manager

REPUTATION MANAGEMENT

Multi-platform review management with sentiment analysis and AI-generated responses.

React

Node.js

Claude



AI Customer Support

CUSTOMER SERVICE

Ticket routing, AI response suggestions, knowledge base, escalation management.

React

Node.js

WebSocket



AI Sales Outreach

SALES AUTOMATION

Personalized outreach generation, sequence management, AI follow-up suggestions.

React

Node.js

Email API



AI Marketing Automation

MARKETING OPS

Campaign management, audience segmentation, A/B testing, optimization recommendations.

React

Node.js

Analytics



AI Social Proof

CONVERSION

Generating testimonials, case studies, and success stories from customer data.

React

Node.js

Claude AI



AI Video Script

VIDEO MARKETING

AI script generation, storyboard creation, social media clip suggestions.

React

Node.js

OpenRouter



AI Proposal Generator

SALES ENABLEMENT

Automated proposals with templates, scope definition, AI content suggestions.

React

Node.js

PDF Export



AI Grant Writer

NON-PROFIT

Opportunity matching, proposal drafting, budget planning, compliance checking.

React

Node.js

Claude AI



AI Contract Analysis

LEGAL TECH

Clause extraction, risk identification, negotiation suggestions, term comparison.

React

Node.js

NLP



AI Recruiter Screener

HR TECH

Resume parsing, skill matching, interview questions, bias-free evaluation scoring.

ReactNode.jsNLP



AI Resume Builder

CAREER SERVICES

Resume optimization, cover letter generation, job matching, application tracking.

ReactNode.jsPDF Parser



AI Onboarding Flow

HR OPERATIONS

Employee onboarding automation with task sequences and progress tracking.

ReactNode.jsWorkflow



AI Meeting Agent

PRODUCTIVITY

Agenda creation, note-taking, action item extraction, automated follow-ups.

ReactNode.jsNLP



AI Knowledge Base

DOCUMENTATION

Automated knowledge base with content organization, search, AI article generation.

ReactNode.jsFull-Text



AI Code Review

DEVELOPER TOOLS

Documentation generation, review suggestions, technical debt identification.

ReactNode.jsAST Parser



AI Personalized Tutor

EDTECH

Adaptive learning with personalized lessons, progress tracking, quiz generation.

ReactNode.jsClaude AI



AI Scheduler

SCHEDULING

Natural language scheduling with voice commands, conflict detection, calendar sync.

ReactNode.jsWeb Speech



AI Menu Digitizer

FOOD TECH

Menu digitization with image analysis, allergen detection, calorie estimation, translation.

ReactTypeScriptClaude Haiku



AI Floor Plan Analyzer

REAL ESTATE TECH

Floor plan analysis with room detection, measurements, furniture placement suggestions.

ReactNode.jsCV



AI Product Photo

E-COMMERCE

Product image optimization with background removal, quality enhancement.

ReactNode.jsImage API



AI Interior Design

DESIGN TECH

Style recommendations, furniture suggestions, color palette generation.

ReactNode.jsOpenRouter

AI Food Flow Order	Restaurant ordering with menu management, order tracking, kitchen display, notifications	React, Node.js, PostgreSQL
AI Recipe Meal Planner	Meal planning with AI recipe generation, grocery lists, nutrition tracking	React, Sequelize, OpenRouter
AI Travel Itinerary	Travel planning with destinations, itinerary generation, booking integration	React, Node.js, PostgreSQL
Beauty & Wellness AI	Salon/spa management with booking, profiles, treatment recommendations	React, Node.js, PostgreSQL
Pet Grooming App	Pet service management with appointments, client management, service tracking	React, Node.js, PostgreSQL
Small Law Firm	Legal practice with case tracking, document management, billing, client portal	React, Node.js, PostgreSQL
Real Estate Agency	Property management with listings, leads, showings, market analysis	React, Node.js, PostgreSQL
Government Contracts	Contract platform with opportunity tracking, proposals, compliance tools	React, Node.js, PostgreSQL
E-Commerce Agent	E-commerce with product management, orders, inventory sync, analytics	React, Node.js, PostgreSQL
AI Voice Agent	Voice-enabled AI with speech recognition, NLU, response generation	React, Node.js, Web Speech
Healthcare Scheduler	Medical appointments with patient management, reminders, insurance verification	React, Node.js, PostgreSQL
Fitness Trainer App	Personal training with workout plans, progress tracking, client communication	React, Node.js, PostgreSQL

TECHNICAL ARCHITECTURE & KEY ACHIEVEMENTS

React 18

FRONTEND

Node.js

BACKEND

PostgreSQL

DATABASE

Claude AI

AI INTEGRATION

9x Performance Improvement

Reduced indexing from 360 to 40 min for 5M products

99.5% Extraction Rate

XQUERY extraction for 700K lines COBOL code

50+ AI Applications

Full-stack production apps across 15+ industries

Enterprise ML Leadership

Transformer-based image curation at Lowe's